

fontc_crater “could not locate the source”: root causes & the path to the correct onboarding commit

Model: Claude Opus 4.8 (1M context) · drafted 2026-06-08 · crater run 2026-04-27 (fontc 5b20924e, google/fonts 755538365478)

1. Executive summary

fontc_crater reports **39 repositories** in the *latest run* as failed before any compiler ran, with reasons of the form “*missing source* ‘...’”. This is the single most actionable category for contributing back to the fontc team: these are **not** fontc compiler bugs — **fontc never started**, because crater could not locate the source file the build config points at. We diagnosed every one against bare-mirror checkouts of the upstream repos at the exact revisions crater used.

The central finding: **every one of these targets is config_is_external: true** — crater reads the *same* google/fonts/ofl/<family>/config.yaml we maintain and inherits the same provenance question (which commit) our upstream_info.md investigations already track. **So this is not a separate project: it is the google/fonts source-metadata enrichment we are already doing.** Fixing the config path or the commit in google/fonts fixes crater on its next run.

2. Method

For each failed repo we read crater’s declared target (crater_targets: upstream URL, pinned rev, config path, config_is_external), located the bare mirror in the local archive (gflib-data/archive/<owner>/<repo>.git), and checked with git cat-file -e / ls-tree whether the expected source exists (a) at crater’s pinned rev, (b) at the commit recorded in our google/fonts METADATA.pb, and (c) at HEAD — plus what sources *do* exist and when the expected one first appeared. The commit dates establish the chronology that distinguishes a stale pin from a genuinely-missing source.

3. Root causes — all 39

Category	n	Why crater can’t locate the source	Contribution / fix
WRONG_PATH	10	Config sources: path doesn’t match the repo layout	Correct the path in the google/fonts override config.yaml
STALE_REV (A)	6	Pinned commit predates the source, but it is at the METADATA commit	Point crater at the METADATA commit (known-good)
STALE_REV (B)	3	Pinned commit and the METADATA commit both predate the buildable source	Forensic onboarding-commit hunt (the upstream_info.md effort)
GENERATED	7	The expected source is produced by a build script, never committed	Commit generated sources under sources/generated/ or add a build step
NOT_MIRRORED	8	Repo not yet in our archive — not diagnosable offline	Mirror into the archive, then re-diagnose
SOURCE_PRESENT	2	Source is present at the pinned rev — crater failed for another reason	Inspect individually (.glyphspackage / .designspace)
ESCAPES_REPO	1	Config path escapes the repo root (./..., mono-repo)	Fix the mono-repo path / config
NO_CONFIG	1	No config file found at all	Add a config.yaml
NO_SOURCES_FIELD	1	Config has no sources field (custom pipeline)	Custom-pipeline family — needs a tailored target
total	39		

4. The unifying insight: crater inherits our config and our provenance gap

Because these targets are external, the contribution decomposes cleanly:

- Path-level failures** (WRONG_PATH, ESCAPES_REPO, NO_CONFIG, NO_SOURCES_FIELD): the sources: path is wrong or absent. Fixing the google/fonts override config.yaml — our existing override workflow — **fixes crater automatically**, no separate crater PR. *Highest-leverage bucket.*
- Commit-level failures** (STALE_REV): the config is fine; crater is building the wrong commit. The fix is to point it at the correct onboarding commit (§5).
- Structural failures** (GENERATED, SOURCE_PRESENT, NOT_MIRRORED): need source-generation handling, individual inspection, or mirroring first.

5. Pointing at the correct onboarding commit

The STALE_REV bucket splits exactly along the provenance question — and that split is where the goal of “*the commit originally used to build the binaries on Google Fonts*” becomes central:

- Sub-case A** — crater pins an *older* commit, but the source is present at our recorded METADATA commit. The correct commit is known and verified; crater just needs it.
- Sub-case B** — crater’s rev *equals* our METADATA commit, and *neither* has the buildable source (it appears only later). Here **google/fonts’ own metadata points at a pre-source commit**, so the onboarding commit is wrong/unknown and must be

recovered by binary↔source comparison. Fixing it benefits both repos.

Family	Sub	crater rev / date	METADATA commit / date	src @META	Correct commit
lalezar	A	c3e0eae242 2016-08-22	238701c424 2017-02-28	yes	238701c424
amethysta	B	10ae36bc06 2016-04-06	10ae36bc06 2016-04-06	no	*unresolved*
jacquesfrancois	A	bc37f476a7 2016-01-10	d34156392b 2018-02-12	yes	d34156392b
jacquesfrancoisshadow	A	90c9f94cc7 2016-01-10	073491c6b1 2018-02-12	yes	073491c6b1
prata	B	db5f3799a4 2016-12-14	db5f3799a4 2016-12-14	no	*unresolved*
intelonemono	A	cec102c389	99e2d6ca17 2024-07-26	yes	99e2d6ca17
notosansarabic	A	133ccaebf9 2022-08-01	6c8320740d 2023-11-07	yes	6c8320740d
raleway	A	7b288c6faa	938ac77022 2020-08-26	yes	938ac77022
castoro	B	58a386a96e 2023-03-15	58a386a96e 2023-03-15	no	*unresolved*

6. The upstream_info.md investigations are the contribution vehicle

The provenance answers crater needs already live (or belong) in our per-family investigation reports. Two worked examples from the sub-case B set show why this is the binding constraint:

- **Jacques François** — upstream_info.md status missing_commit: “the original commit cannot be reliably determined... needs investigation to identify which upstream commit corresponds to the font currently in google/fonts.” Crater pinned the 2016 initial commit (prebuilt TTF only, no sources/); the modern source exists at the 2018 d341563 regeneration.
- **Amethysta** — upstream_info.md already flags it “WON’T FIX (unreproducible)”: the recorded commit 10ae36bc (2016) carries only the prebuilt TTF; the source first appears in d3cd4ca (“Release v1.002”), yet the shipped binary is v1.003 (2017). The artifact on Google Fonts does not correspond to a buildable commit in the repo.

So crater’s “could not locate the source” is, for the hardest cases, the same unresolved provenance our enrichment effort is tracking. Each resolved onboarding commit is a single fix that lands in upstream_info.md, corrects METADATA.pb, and unblocks crater at once.

7. Proposed contribution path

1. **Path fixes (the WRONG_PATH + structural-config bucket).** Correct each google/fonts override config.yaml sources: path (Appendix B); crater inherits it. Fastest, auto-propagates.
2. **Stale-rev sub-case A.** Supply the verified METADATA commit to crater’s target list (confirm first whether crater resolves its rev from METADATA or from its own sources.json).
3. **Stale-rev sub-case B.** Run the binary↔source forensic hunt; record the onboarding commit in upstream_info.md + METADATA.pb; some (amethysta) may stay unreproducible and should be reported to crater as such.
4. **Generated sources.** Adopt the sources/generated/ convention so both we and crater can build them.
5. **Not-mirrored.** Mirror into the archive, then re-run this diagnosis.

Appendix A — per-repository diagnosis (all 39)

Repo	Family	Category	Missing source (crater config)	crater rev
alexseiva/arsenal	arsenal	WRONG_PATH	Arsenal-Italic.glyphs	878af08407
googlefonts/sawarabi-mincho	sawarabimincho	WRONG_PATH	SawarabiMincho.glyphs	8bd1525717
gue3bara/cairo	cairo	WRONG_PATH	CairoPlay.glyphs	73d16933c6
itfoundry/kumar	kumaroneoutline	WRONG_PATH	Kumar One.glyphs	3192a79a79
lipiraval/mogra	mogra	WRONG_PATH	Mogra.glyphs	048039d237
m4rcle/istok-web	istokweb	WRONG_PATH	IstokWeb-Italic.glyphs	f995ade617
mooniak/abhaya-libre-font	abhayalibre	WRONG_PATH	sources/glyphs/Abhaya-Masters.glyphs	f53da70786
omnibus-type/barrio	barrio	WRONG_PATH	sources/Barrio.glyphs	8f33bf10cb
omnibus-type/pragatinarrow	pragatinarrow	WRONG_PATH	PragatiNarrow.glyphs	829be323c4
weiweihuanguang/fragment-mono	fragmentmonosc	WRONG_PATH	Fragment-Mono.glyphs	3ff027831f
bornaiz/lalezar	lalezar	STALE_REV (A)	sources/Lalezar.glyphs	c3e0eae242
cyrealtype/jacques-francois	jacquesfrancois	STALE_REV (A)	sources/JacquesFrancois.glyphs	bc37f476a7
cyrealtype/jacques-francois-shadow	jacquesfrancoisshadow	STALE_REV (A)	sources/JacquesFrancoisShadow.glyphs	90c9f94cc7
intel/intel-one-mono	intelonemono	STALE_REV (A)	sources/masters/IntelOneMono-Roman.des	cec102c389
notofonts/arabic	notosansarabic	STALE_REV (A)	sources/NotoNaskhArabicUI.glyphspackag	133ccaebf9
theleagueof/raleway	raleway	STALE_REV (A)	sources/Raleway-Italic.designspace	7b288c6faa
cyrealtype/amethysta	amethysta	STALE_REV (B)	sources/Amethysta-Regular.glyphs	10ae36bc06
cyrealtype/prata	prata	STALE_REV (B)	sources/Prata.glyphs	db5f3799a4
tirotypeworks/castoro	castoro	STALE_REV (B)	source/Castoro-Italic.designspace	58a386a96e
aaronbell/lxgwmargergothic	lxgwmargergothic	GENERATED	temp/LXGWMarkerGothic-Regular.ufo	fe83570074
cadsondemak/sriracha	sriracha	GENERATED	Sriracha-Regular.ufo	6c6cf92ed8
cathschmidt/yatra-one	yatraone	GENERATED	YatraOne_0.ufo	b991e49f27
moonlitowen/cactusserif	cactusclassicalserif	GENERATED	temp/CactusClassicalSerif-Regular.ufo	a267f9f320
moonlitowen/chocolatesans	chocolateclassicalsans	GENERATED	temp/ChocolateClassicalSans-Regular.uf	624ecb8064
moonlitowen/thenkhung	uoqmunthenkhung	GENERATED	temp/UoqMunThenKhung-Regular.ufo	cdf0805fd0
sorkintype/trykker	trykker	GENERATED	Trykker	5226cb0750
anoxic/neuton	neuton	NOT_MIRRORED	NL.ufo	b376055d27
gitlab.com/smc/fonts/manjari		NOT_MIRRORED	sources/Manjari-Bold.ufo	8948773e57
googlefonts/cousine	cousine	NOT_MIRRORED	sources/Cousine-BoldItalic.glyphs	7f897dbd87
googlefonts/noto-fonts		NOT_MIRRORED	sources/NotoSansGurmukhi.glyphs	090cc7e2cf
googlefonts/tinos	tinos	NOT_MIRRORED	sources/Tinos-BoldItalic.glyphs	aaf68d53c2
loudifier/comic-relief	comicrorelief	NOT_MIRRORED	ComicRelief-Bold.ufo	856315f5a4
notofonts/duployan	notosansduployan	NOT_MIRRORED	NotoSansDuployan.glyphs	9626869cc8
vernnobile/oxygenfont	oxygenmono	NOT_MIRRORED	Oxygen-Regular.ufo	62db0ebe34
aliftype/amiri	amiri	SOURCE_PRESENT	sources/Amiri.glyphspackag	480bb746e9
notofonts/nyiakang-puachue-hmong	notoserifnphmong	SOURCE_PRESENT	sources/NotoSerifNPHmong.designspace	2c945bb9c3
xotypeco/big_shoulders	bigshouldersdisplaysc	ESCAPES_REPO	../Big-Shoulders/sources/BigShoulders.	0b3d09a868
googlefonts/dm-fonts	dmsans	NO_CONFIG	no config file was found	027cea4e4f
petrvanblokland/typetr-bitcount	bitcountprosingleink	NO_SOURCES_FIELD	missing field sources`	653fc48a72

Appendix B — `WRONG_PATH` source-path corrections

The config points one place; the repo keeps the source another. Correct the `sources:` entry in each google/fonts override `config.yaml` (crater then inherits it).

Family	config <code>sources:</code> says	repo actually has
abhayalibre	<code>sources/glyphs/Abhaya-Masters.glyphs</code>	<code>documentation/development/03-08-15/Abhaya-Masters-09.glyphs</code>
arsenal	<code>Arsenal-Italic.glyphs</code>	<code>sources/Arsenal-Italic.glyphs</code>
barrio	<code>sources/Barrio.glyphs</code>	<code>Barriecito/sources/Backup/1138_Barriecito.glyphs</code>
cairo	<code>CairoPlay.glyphs</code>	<code>sources/Archive/Cairo-italic.glyphs</code>
fragmentmonosc	<code>Fragment-Mono.glyphs</code>	<code>sources/Fragment-Mono.glyphs</code>
istokweb	<code>IstokWeb-Italic.glyphs</code>	<code>old/v.1.0.3/sources/SDF-PS/Istok-Bold.sfd</code>
kumaroneoutline	<code>Kumar One.glyphs</code>	<code>masters/Kumar One.glyphs</code>
mogra	<code>Mogra.glyphs</code>	<code>sources/Mogra.glyphs</code>
pragatinarrow	<code>PragatiNarrow.glyphs</code>	<code>SRC/PragatiNarrow.glyphs</code>
sawarabimincho	<code>SawarabiMincho.glyphs</code>	<code>fonts/gothic/SawarabiGothic-Medium.sfd</code>